

# **Information Technologies for Industrial Engineers**

# Backend Application with Database

# Setup Database

```
docker run -d --name my-postgres -e POSTGRES_PASSWORD=1234 -v  
pgdata:/var/lib/postgresql -p 5432:5432 postgres:latest
```

Note that this command specifies the password of **1234** for the database.  
You can change it to your own password.

# Prepare Database

In Dbeaver

```
CREATE DATABASE app_db;
```

```
CREATE TABLE IF NOT EXISTS
  todos (
    id serial PRIMARY KEY,
    title VARCHAR(50) NOT NULL,
    completed BOOLEAN NOT NULL DEFAULT FALSE,
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
  );
```

Note the database name is `app_db` and the table name is `todos`.

# Backend Starter

- `git clone https://github.com/it-for-ie-69/express-template.git express-db`
- `cd express-db`
- `git init` *(if you want to remove the previous git history)*
- `pnpm i pg dotenv`
- `pnpm i -D @types/pg`
- Create `.env` and paste the content from [here](#).
  - Change the database password to your own password.

# Starter Code

- `index.js`

# Example Code

- `./src/db.ts`
- `./src/index.ts`